

Notes on the product of exponentials formula, MLS Chapter 3, Section 2.2

In the middle of p. 86, MLS argues that, after having kept joint 2 fixed and rotated joint 1 through θ_1 , if joint 1 is next kept fixed and joint 2 is to be actuated through an angle θ_2 , the transformation $e^{\hat{\xi}_2\theta}$ will no longer be valid. This is because the axis of rotation of joint 2 has moved, due to the first rotation through θ_1 (that is, using $e^{\hat{\xi}_2\theta}$ will violate the statement in the last sentence of the paragraph immediately following eq. (3.2), that the screw motion for each joint at the $\theta_1 = \theta_2 = 0$ configuration of the manipulator should be evaluated). They then claim that the correct twist to use, instead, is ξ'_2 , where

$$\xi'_2 = \text{Ad}_{e^{\hat{\xi}_1\theta_1}} \xi_2 = \text{Ad}_g \xi_2$$

where

$$g = e^{\hat{\xi}_1\theta_1}.$$

This transformation with the adjoint matrix is not something new; it was introduced in equations (2.57) and (2.58), p. 55, to transform the body velocity to the spatial velocity. Take note, however, of the last paragraph of MLS p. 59, and specifically eq. (2.64). This equation is not well proven on p. 59, but this is once again addressed in problem 13 (b) on p. 77. The proof of eq. (2.64) or the solution of problem 13 (b) is included as a handwritten note at the back of this document.

Lemma 2.13 p. 56: states that the twist with coordinates $\text{Ad}_g \xi_2 \in \mathbb{R}^6$ has the corresponding form $g\hat{\xi}_2g^{-1} \in \mathfrak{se}(3)$. Based on this, we can write

$$e^{\hat{\xi}'_2\theta_2} = e^{g\hat{\xi}_2g^{-1}\theta_2}.$$

In the March 9, 2026 lecture, we have proven the identity used in eq. (2.35) (p. 42; see slides on Moodle) and which is now also used here to simplify the above equation to get the result shown on MLS p. 86:

$$e^{\hat{\xi}'_2\theta_2} = e^{g\hat{\xi}_2g^{-1}\theta_2} = g e^{\hat{\xi}_2\theta_2} g^{-1} = e^{\hat{\xi}_1\theta_1} \left(e^{\hat{\xi}_2\theta_2} \right) e^{-\hat{\xi}_1\theta_1}$$

Homework 13(b) p. 77, MLS:

$$A_d g = \begin{bmatrix} R & \hat{P}R \\ 0 & R \end{bmatrix} \text{ where } g = \begin{bmatrix} R & P \\ 0 & I \end{bmatrix}$$

(eq. (2.58) p. 55)

$$\xi_a = \begin{bmatrix} -\hat{\omega}_a q_a + h\omega_a \\ \omega_a \end{bmatrix}$$

$$\xi_a' = \begin{bmatrix} -\hat{\omega}_a' q_a' + h\omega_a' \\ \omega_a' \end{bmatrix}$$

$$\begin{bmatrix} q_a' \\ 1 \end{bmatrix} = \begin{bmatrix} R & P \\ 0 & I \end{bmatrix} \begin{bmatrix} q_a \\ 1 \end{bmatrix} \Rightarrow q_a' = Rq_a + P$$

$$\omega_a' = R\omega_a$$

From eq. (2.7) pg 26: $\hat{\omega}_a' = R\hat{\omega}_a R^T$

Thus,

$$\xi_a' = \begin{bmatrix} -R\hat{\omega}_a q_a - \hat{\omega}_a' P + hR\omega_a \\ R\omega_a \end{bmatrix}$$

On the other hand,

$$\begin{aligned} \text{Ad}_g \xi_a &= \begin{bmatrix} R & \hat{p}R \\ 0 & R \end{bmatrix} \begin{bmatrix} -\hat{\omega}_a q_a + h\omega_a \\ \omega_a \end{bmatrix} \\ &= \begin{bmatrix} -R\hat{\omega}_a q_a + hR\omega_a + \hat{p}R\omega_a \\ R\omega_a \end{bmatrix} \end{aligned}$$

$$\xi_a' = \begin{bmatrix} R\hat{\omega}_a q_a - \hat{\omega}_a' p + hR\omega_a \\ R\omega_a \end{bmatrix}$$

$$= \begin{bmatrix} -R\hat{\omega}_a q_a + \hat{p}\omega_a' + hR\omega_a \\ R\omega_a \end{bmatrix}$$

$$= \begin{bmatrix} -R\hat{\omega}_a q_a + \hat{p}R\omega_a + hR\omega_a \\ R\omega_a \end{bmatrix}$$

$$= \text{Ad}_g \xi_a$$